POLS 641
Introductory Political Analysis
Class 10: Regression Assumptions and Interactions

Dr. Colin Kuehl

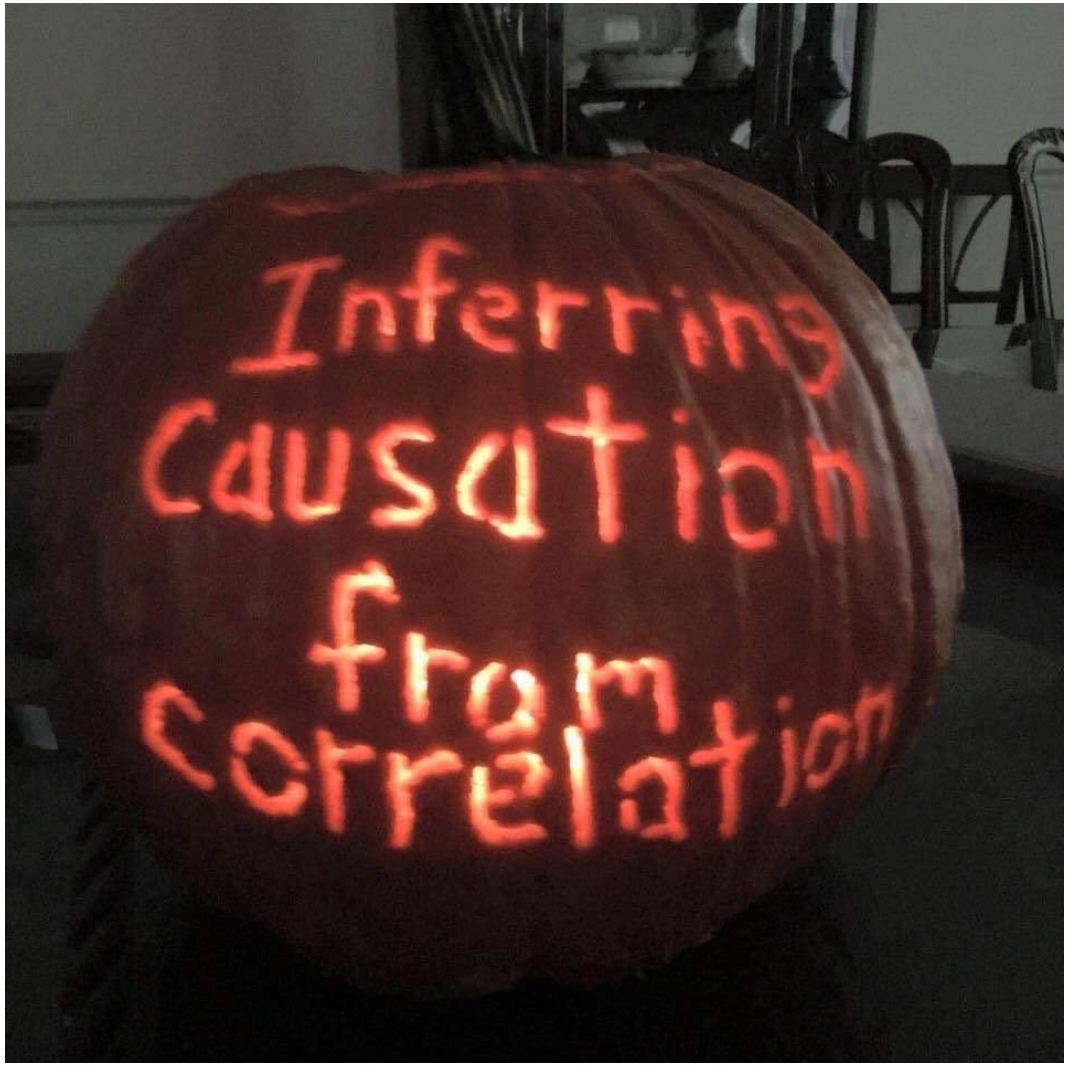
Northern Illinois University

November 5, 2026

Agenda

- Regression Review
- Interaction Terms
- Regression Assumptions and Diagnosing Issues
  - Linearity
  - Outliers
  - MultiCollinearity
  - Heteroskedasticity
  - Autocorrelation

Truly Scary



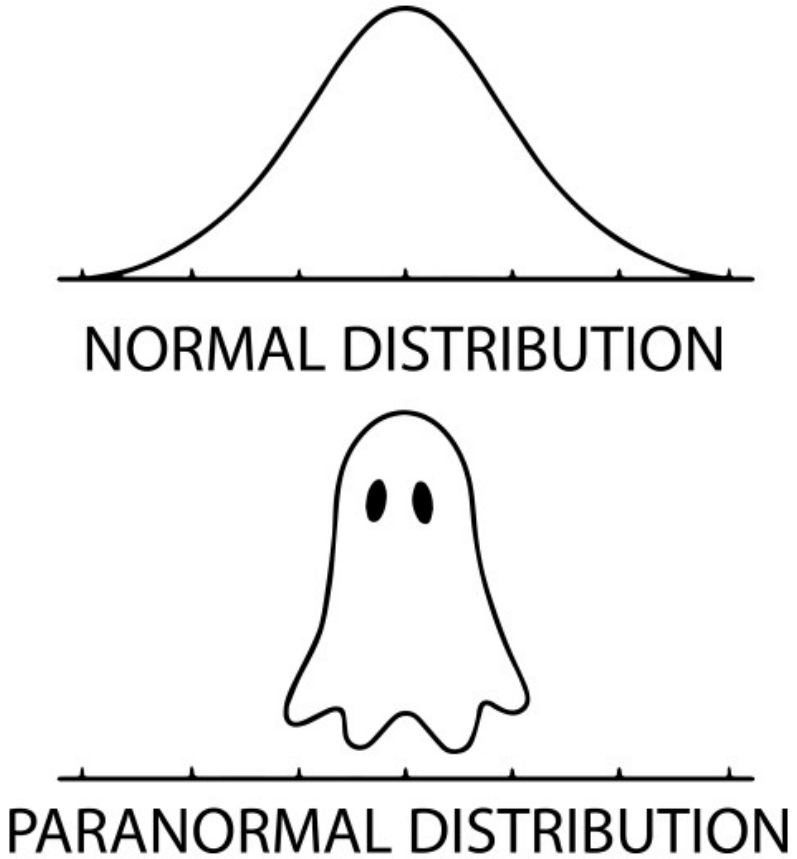
Truly Terrifying



Keeps me up at night



Reviewing Types of Distributions



Course Logistics

Upcoming Schedule

- Today: Model Specification combined with R
- 11-10/12 Causal Inference & Experiments
- **11-17/19 Review & Midterm**
- 11-24 Thanksgiving Week - No Class
- 12-1 Mixed Methods
- 12- 3 Research Presentations
- 12-13 Final Project Due

Midterm

- Paper Portion - Nov 19th - 60 Minutes
  - Multiple-choice, matching, short answer questions
  - Emphasis on understanding the logic of research and being able to "talk like a social scientist"
  - No notes, no computer, no friends
  - Bring a pen . . . . and a positive attitude
- Computer portion - Nov 19th or 20th - 60 Minute Block
  - Mini-problem set format (Rmarkdown with questions and data)
  - You are free to use old notes, google, etc, but not your classmates, "the guru," other humans, or AI
  - If you are a clueless coder time may be an issue. If you're competent(which all of you are) it should not be.
- In total the midterm is worth 10% of your grade. This is equivalent to two problem sets and should reflect a similar level of effort.
- Learning this now will serve you throughout your career

Upcoming Dates and Deadlines

- 11/4 - Datacamp
- 11/11 - Pset # 5 Due
- 12/3 - *Final Project Presentations*
- 12/13 *Final Project due*

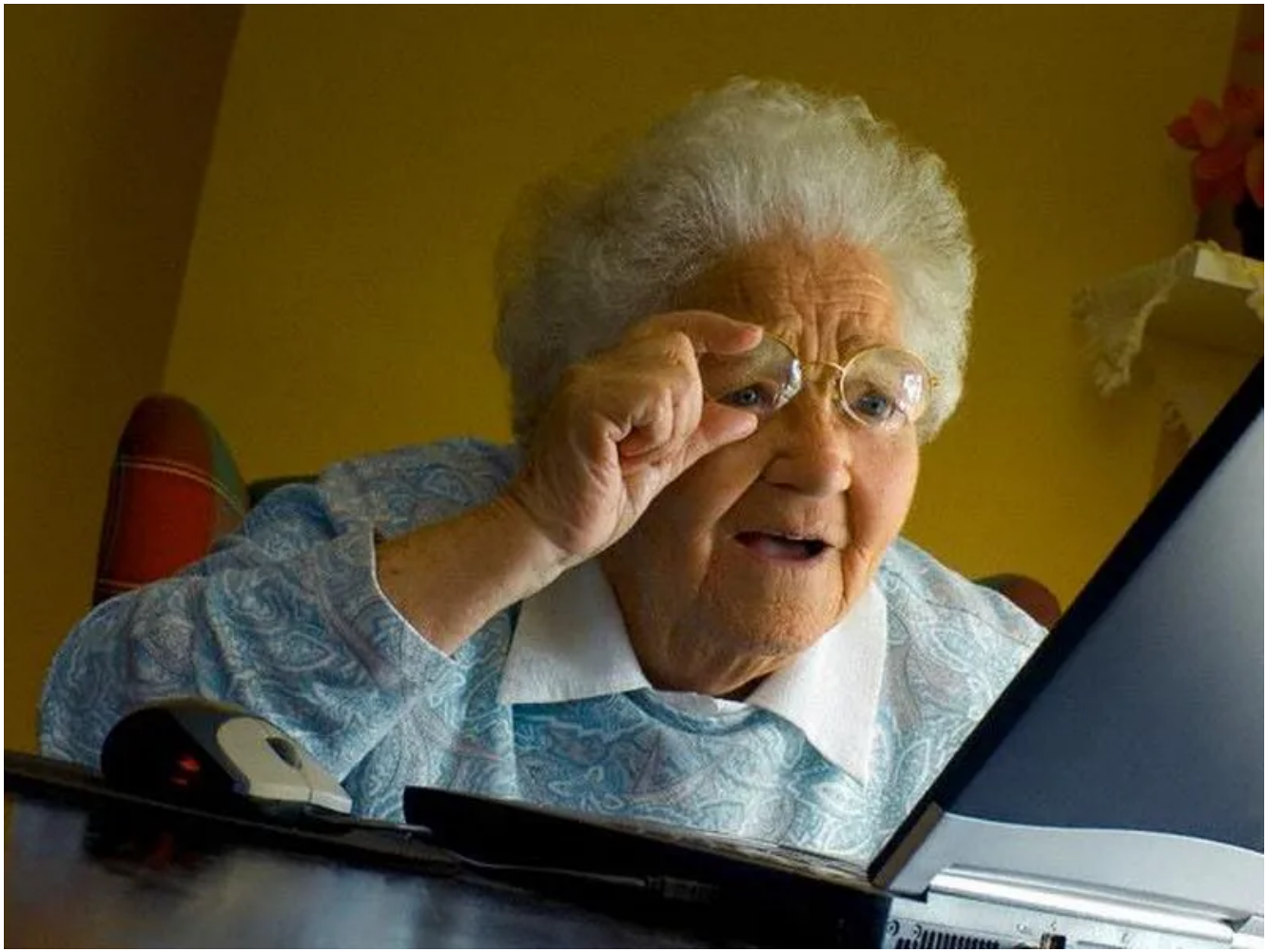
Datacamp

- Take a screenshot when completed and send to me
- 1 Due 11/18

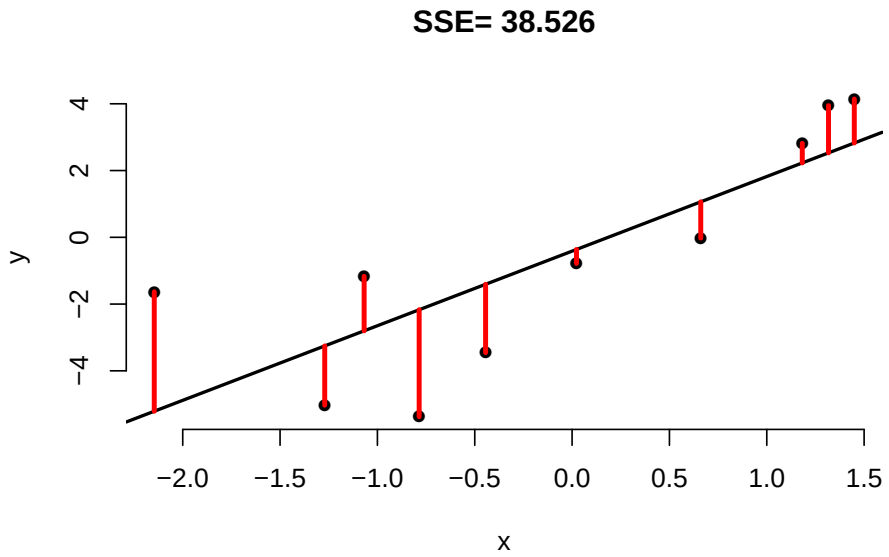
Suggestions

- Data Visualization with GGplot2
- Analyzing Survey Data in R
- Factor Analysis in R
- Experimental Design in R
- Reporting with R Markdown
- Data Manipulation with dplyr
- Foundations of Inference in R
- . . . and many more.

Today: Transformations and Regression Assumptions



Best Fitting Line



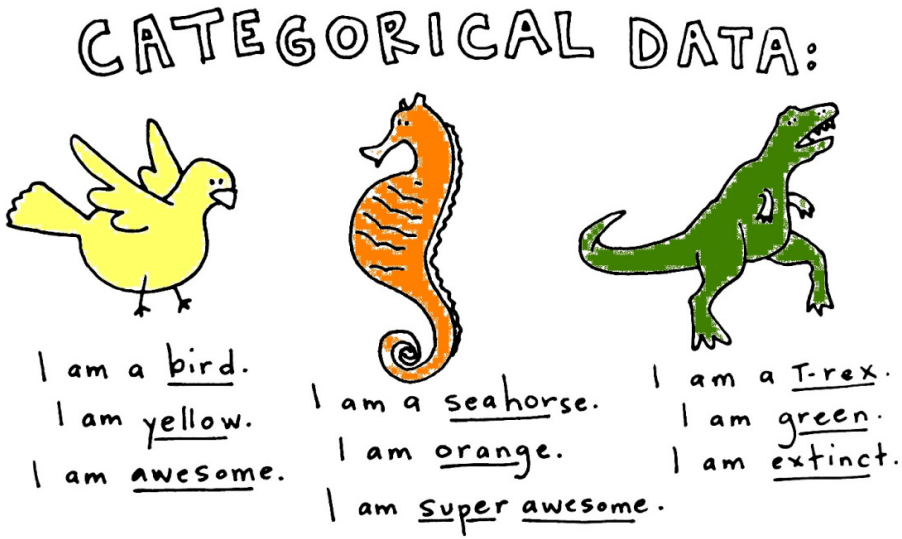
- The residual is the distance between the line and any given point in our data.
- The SSE takes those residuals, squares them, and adds them up.
- The regression shows us the best fitting line in terms of sum of squared errors (SSE).

Review: Binary variables

- Dummy variable - a variable with two distinct categories
- Either on or off, yes/no, war/peace etc.
- take a value of 1(on) or 0 (off)
- Dichotomous, binary, dummy - same thing

Categorical Data

- It is descriptive: it describes how the world is. Often, categorical data comes from qualitative research



- Categorical data can be ordinal or nominal. What does this mean? Ordinal can be ordered (low, medium high) versus nominal cannot be ordered (majors: political science, economics, sociology).

Including Categorical Variables in your Regression

- We add categories to our regression model, but leave out our reference category"

$$Y_i = \beta_0 + \beta_1 Category1_i + \beta_2 Category2_i + \epsilon_i$$

(Note: Depending on data structure in R this could be one factor variable or multiple dummy variables)

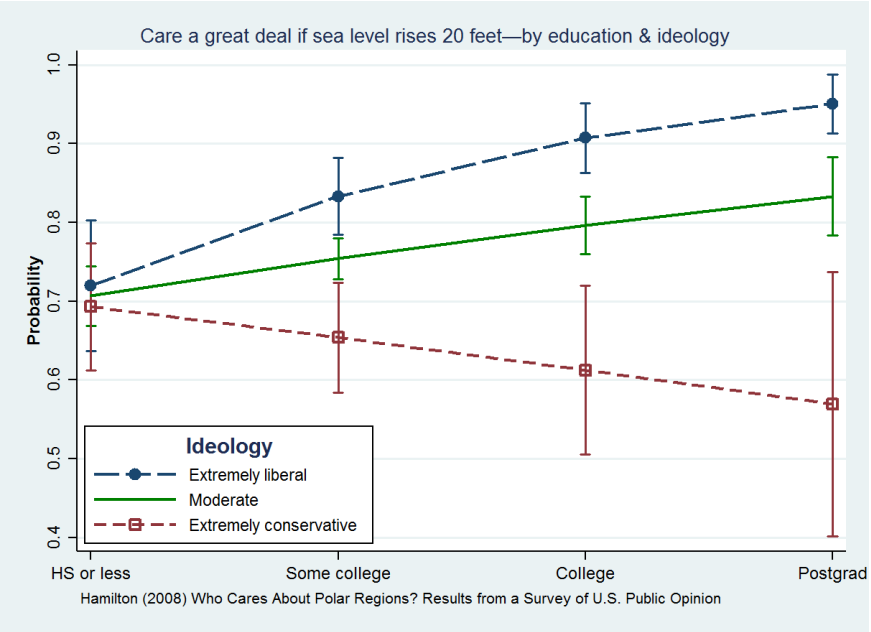
- Interpretation of p-value is the same
- Size of coefficient is interpreted as the difference in dependent between that category and the reference category
- Like a dummy, this has the effect of shifting the Y-intercept

Interaction Terms

Interaction Terms

Interaction effects occur when the effect of one variable depends on the value of another variable.
For example:

- we think the effect of work experience is different more men than women
- The impact of education is dependent on ideology



Interaction Terms

Mathematically this simply means multiplying the two terms.

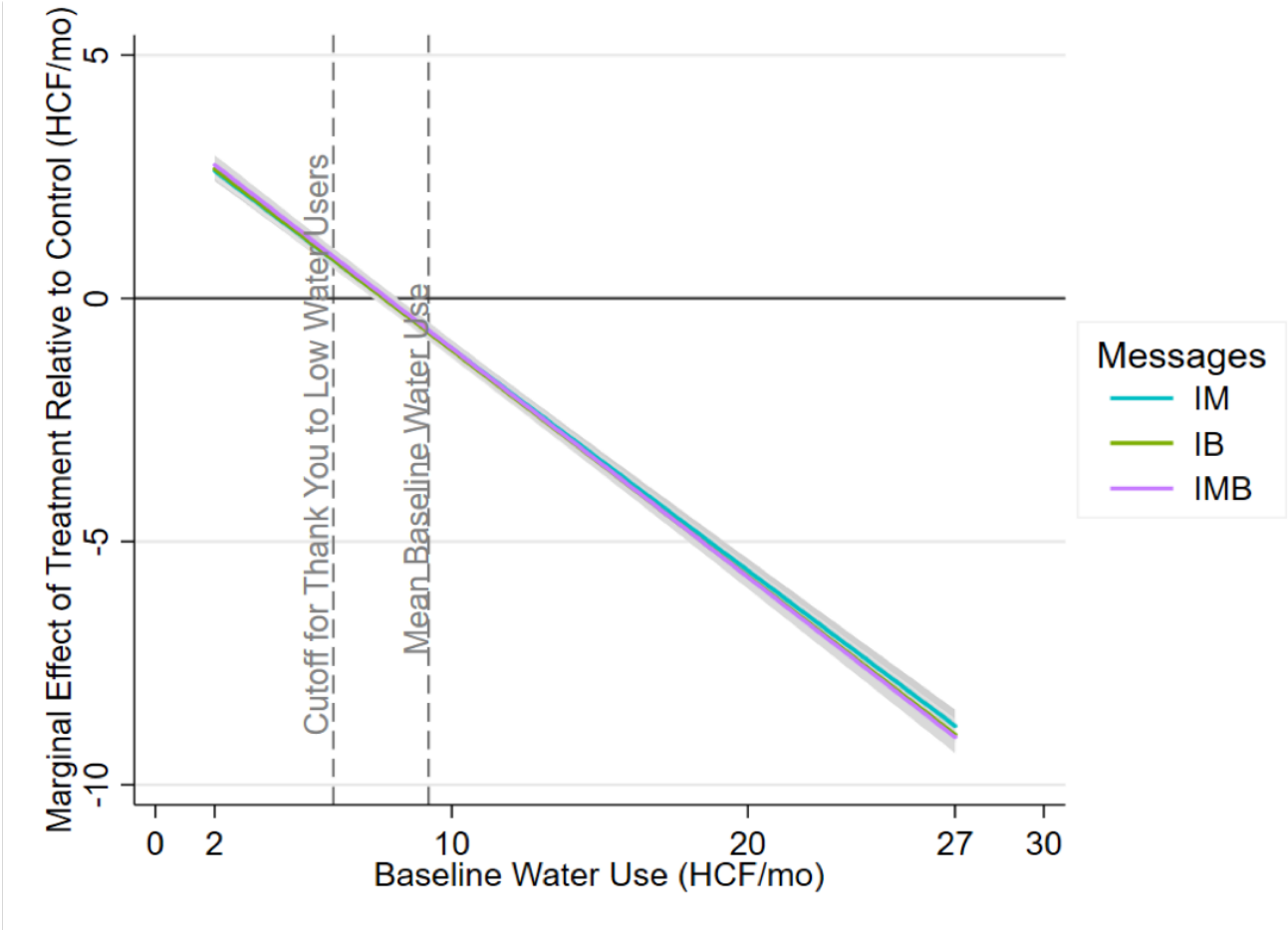
$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i + \epsilon_i$$

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i + \beta_1 X_i * \beta_2 X_i \epsilon_i$$

Best practice is to always include the main effects in the model as well as the interaction term(R automatically does this for you)

Interaction Terms

Example from my work: The impact of different of treatment at different baseline water usage levels



Interaction Terms

Interaction: When the effect of one independent variable differs based on the level or magnitude of another independent variable
In other words, the effect of one variable is conditional on the effect of another.

Interaction Terms

Our core model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \epsilon_i$$

Interaction model:

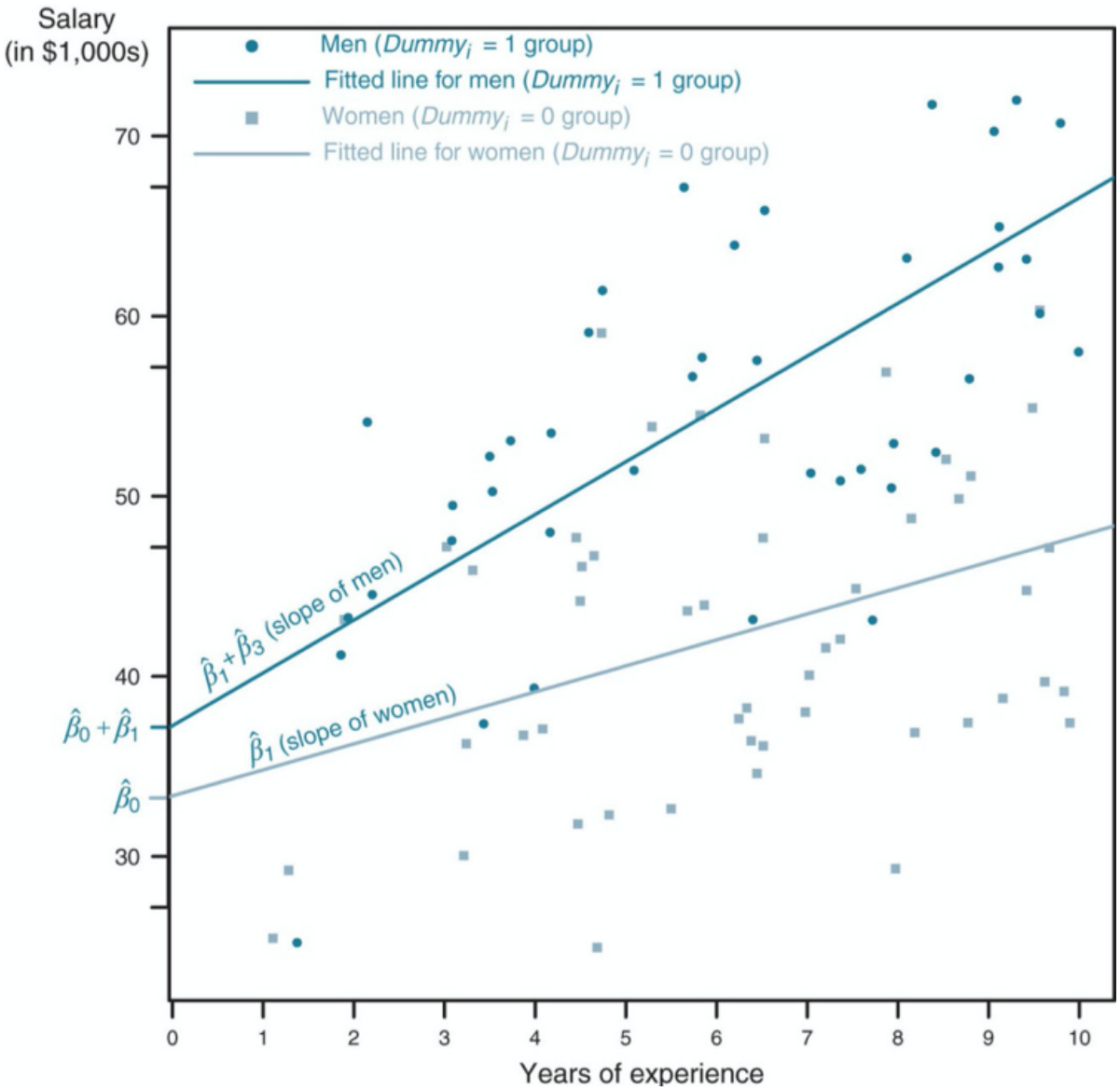
$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_1 X_i * \beta_2 Z_i + \epsilon_i$$

Interaction model:

$$Salary = \beta_0 + \beta_1 Gender + \beta_2 Experience + \beta_1 Gender * \beta_2 Experience + \epsilon_i$$

ie the effect of more experience is conditional on an individual's gender

Interaction terms



What could go wrong?

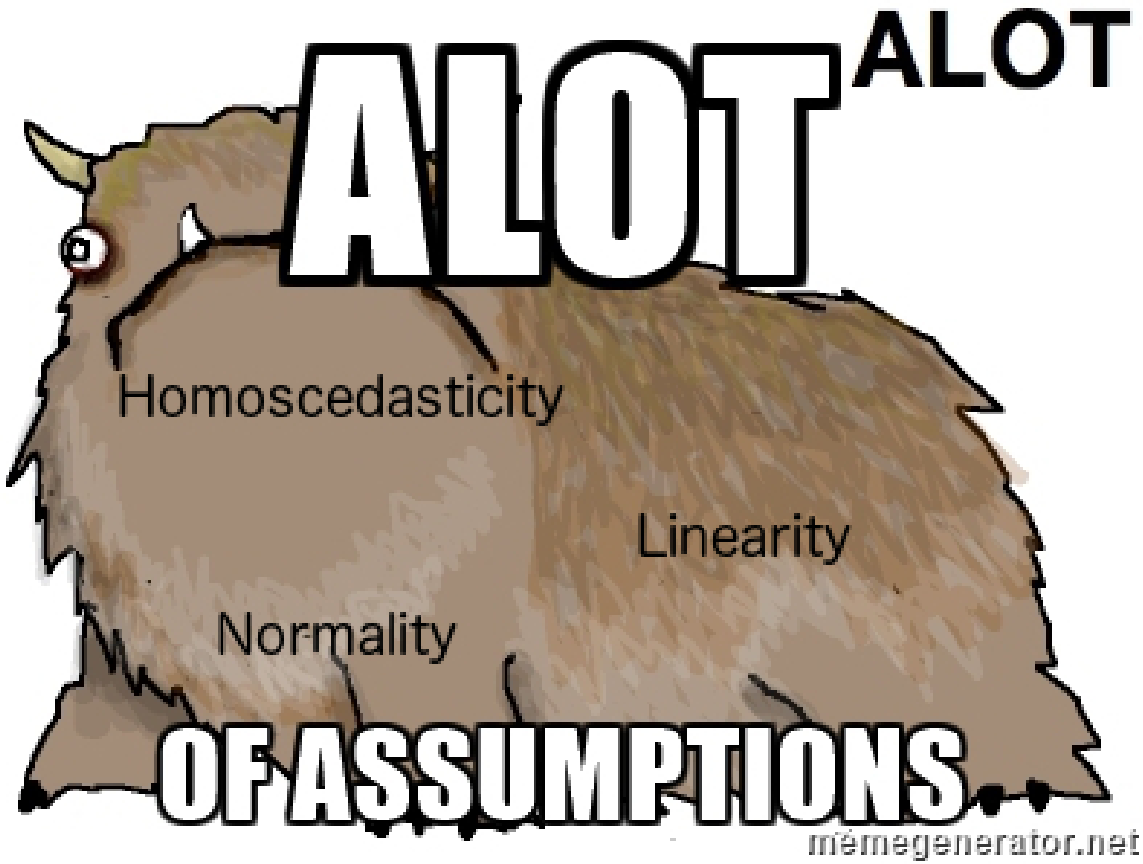
Regression Warnings



Regression Warnings



Regression is a very powerful tool when used correctly
However, when used incorrectly it can be dangerous


ALOT
ALOT
Homoscedasticity
Linearity
Normality
OF ASSUMPTIONS
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Issue: Measurement

- A regression is only as good as the data it comes from
- Operationalization of variables is key(must be justified)
- Variables should be valid and reliable estimators of our estimand

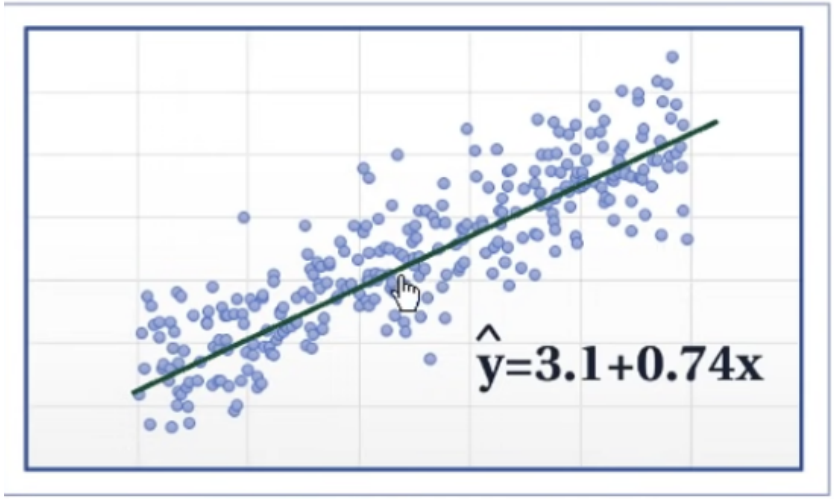
- solution:
  - Find similar variables and demonstrate convergent validity
  - Run models with alternative measures to demonstrate robustness

Issue: Model Specification

- Which independent variables should be included in the regression
- Should be based on theory
- Most important component of regression analysis.

How Regression Goes Wrong

$$Y = \beta_0 + \beta_1 X + \varepsilon$$



	Coef	*Std Err*	*t Stat*	*P-value*
Intercept	3.0605	1.4024	2.1823	0.02909
X	0.7393	0.2308	3.20339	0.00136

- Biased Coefficients
- Faulty Standard Errors(p-values)

Bias in Estimates

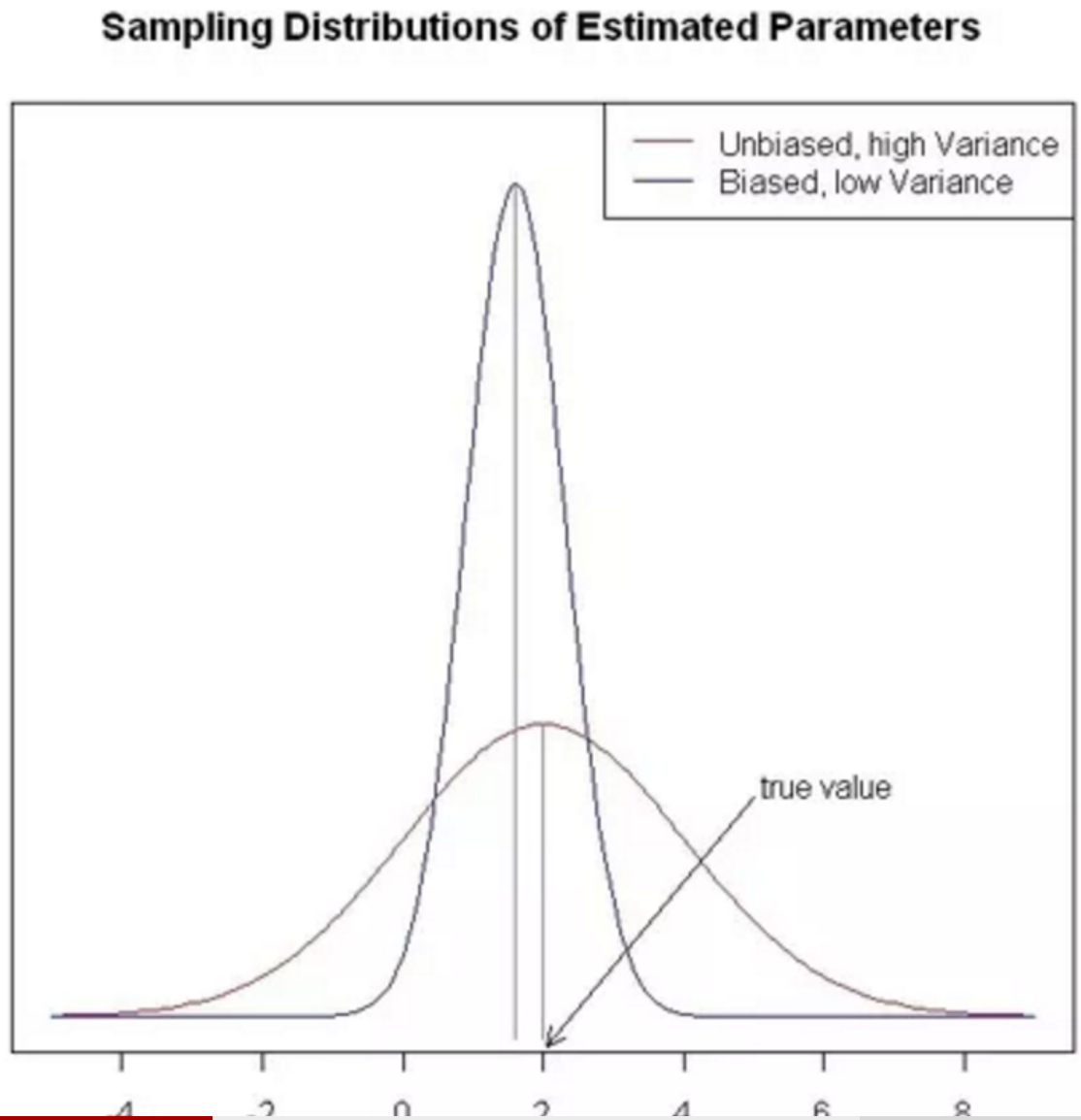
- What is bias?
  - Unbiased estimate: On average, our estimate is equal to the true parameter, $\hat{\beta}_1 = \beta_1$
  - Biased estimate: Our coefficient is systematically wrong, either too high or two low than the true parameter, $\hat{\beta}_1 > \beta_1$
  - *Biased = the slope is wrong*

Omitted Variable Bias

- OLS does not necessarily create unbiased estimates. Why?

- Omitted variable bias: this is a specific form of endogeneity.
  - X is correlated with something else that influences Y; the error term is correlated with Y.
  - This is often the reason why, if you change the model specification, your estimates change. We say, or model is not robust to alternative specifications.
  - Therefore, our model is missing a key confounder, and we haven't estimated a causal relationship.
  - Theoretically, you could include this confounder in your model (multi-variable regression) but this is often hard to do: maybe you don't know what the confounder is, or you don't have data on it.

A Picture of a Biased & Unbiased Estimator



Key Points on Bias + OLS

REMEMBER THIS

1. The distribution of an unbiased estimator is centered at the true value, β_1.
2. The OLS estimator $\hat{\beta}_1$ is an unbiased estimator of β_1 if X and ϵ are not correlated.
3. If X and ϵ are correlated, the expected value of $\hat{\beta}_1$ is $\beta_1 + \text{corr}(X, \epsilon)\frac{\sigma_\epsilon}{\sigma_X}$.

So, how do we get around this bias? Exogeneity!

Consistency in Estimates

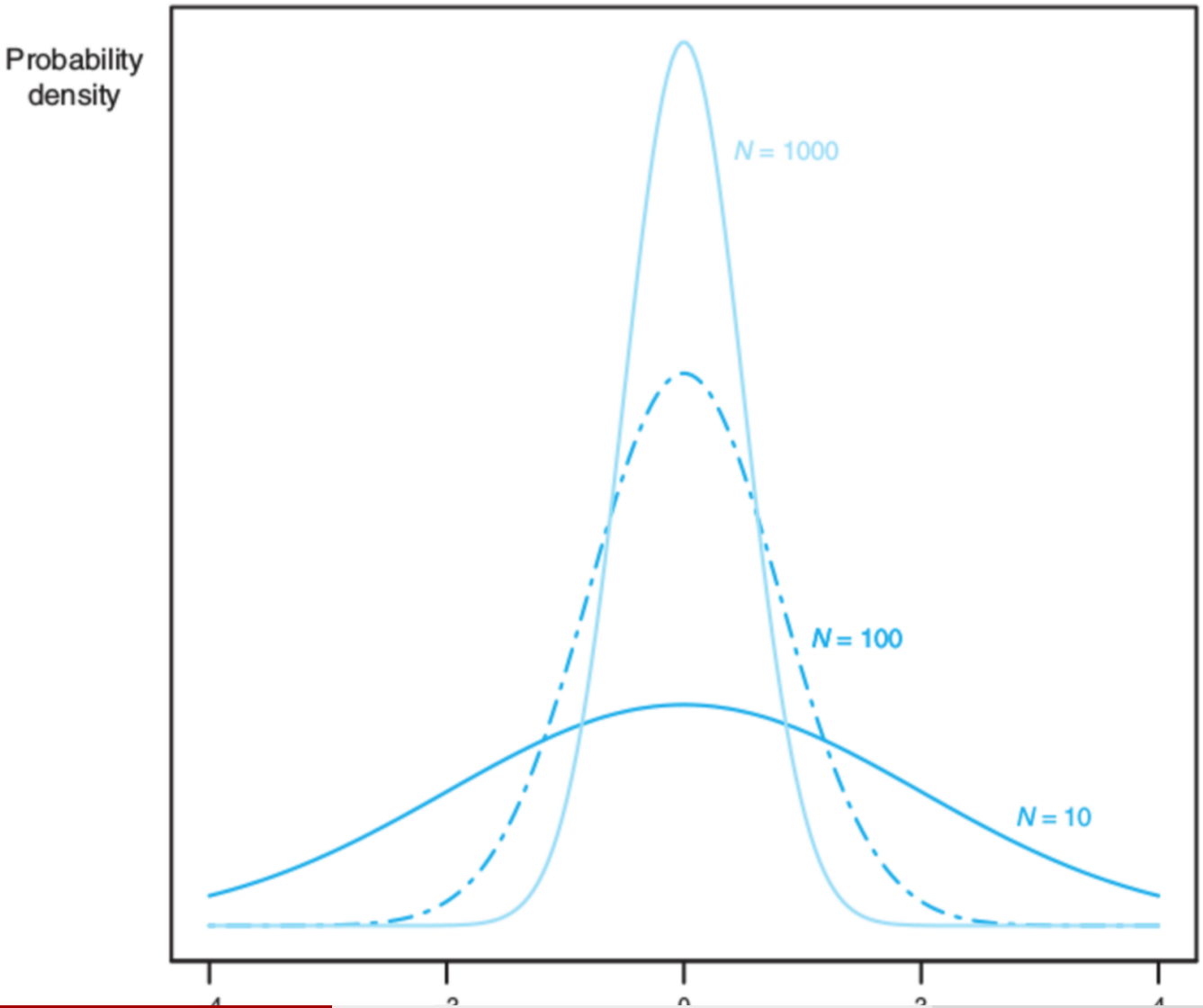
- Consistency:

  As our sample size (N) grows, the estimate converges on the true parameter, $\hat{\beta}_1 \xrightarrow{p} \beta_1$

- OLS estimators are consistent.

- But remember, your estimate will only converge on the true parameter (be consistent) if there is no omitted variable bias in your regression model.

A Picture of Consistency



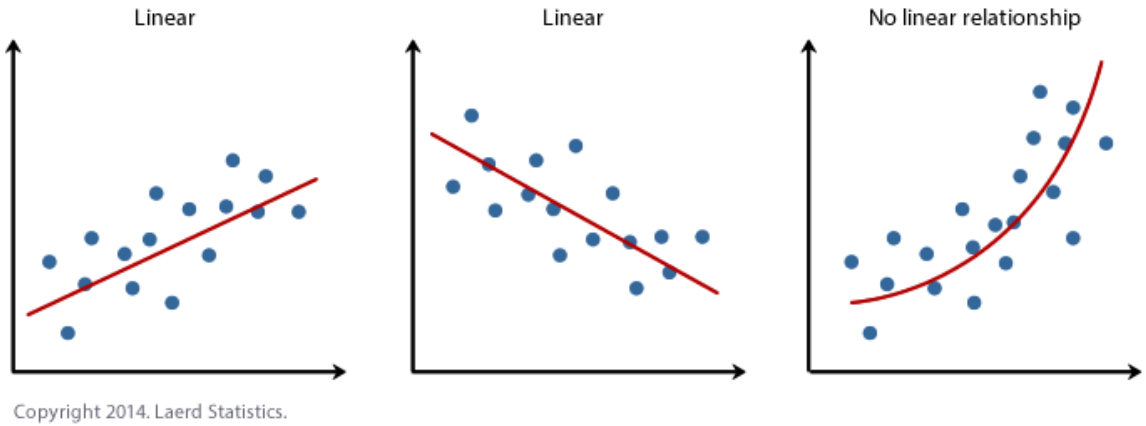
Regression Warnings

Assumptions of the basic OLS model

1. Continuous DV (More on this later or in 642)
2. IV – continuous, dummy, categorical
3. Linear relationship
4. No significant outliers
5. Independence of observations
6. Homoskedasticity
7. Residuals (errors) of regression line normally distributed

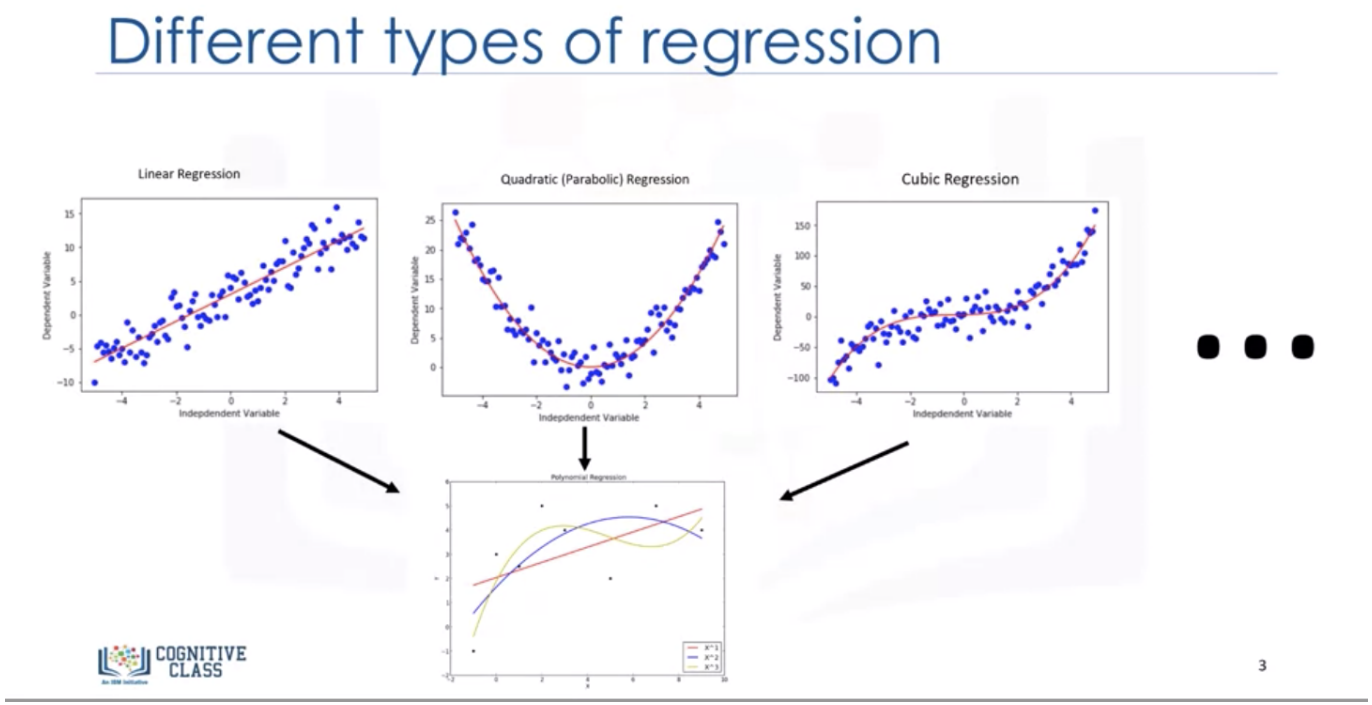
These will be taught in more detail in 642

Non-Linear Models



OLS assumes the effect of X on Y is consistent across different values of X

Non-Linear Models



This fits within our broader understanding of model fit.
Again, this is why always good idea to create a scatter plot

Outliers

- Reminder, what is an outlier? An observation that is extremely difference from the rest of the observations in the sample. “One of these things is not like the others."
- Reminder, what does an outlier do to our estimate of the mean? It drags it towards the outlier. The mean is sensitive to outliers.
- Intuitively, then, what would outliers do to our regression estimates?
  It would also drag our estimate of the slope, $\hat{\beta}_1$ towards the outlier.

A Picture of an Outlier & Regression

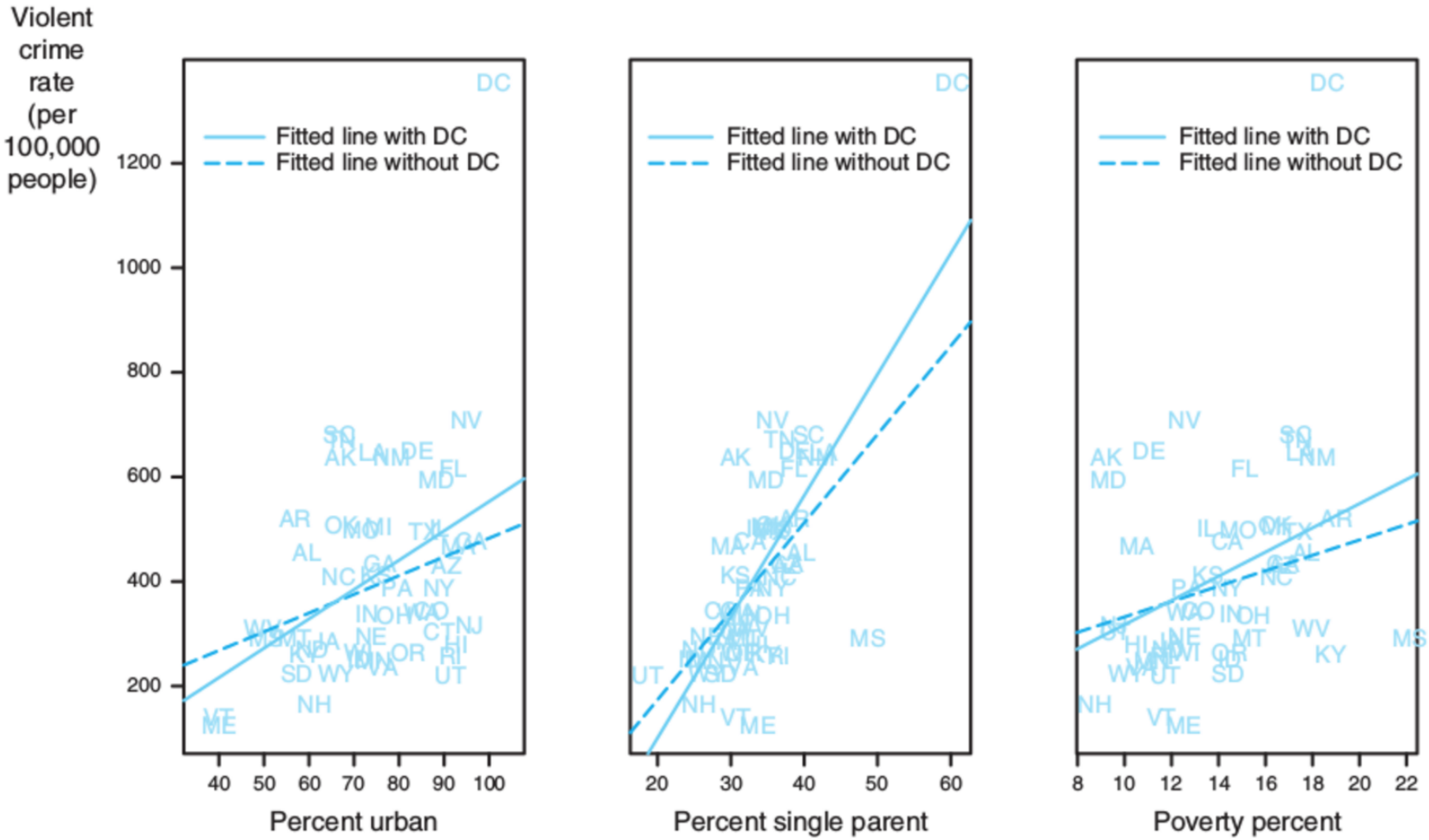


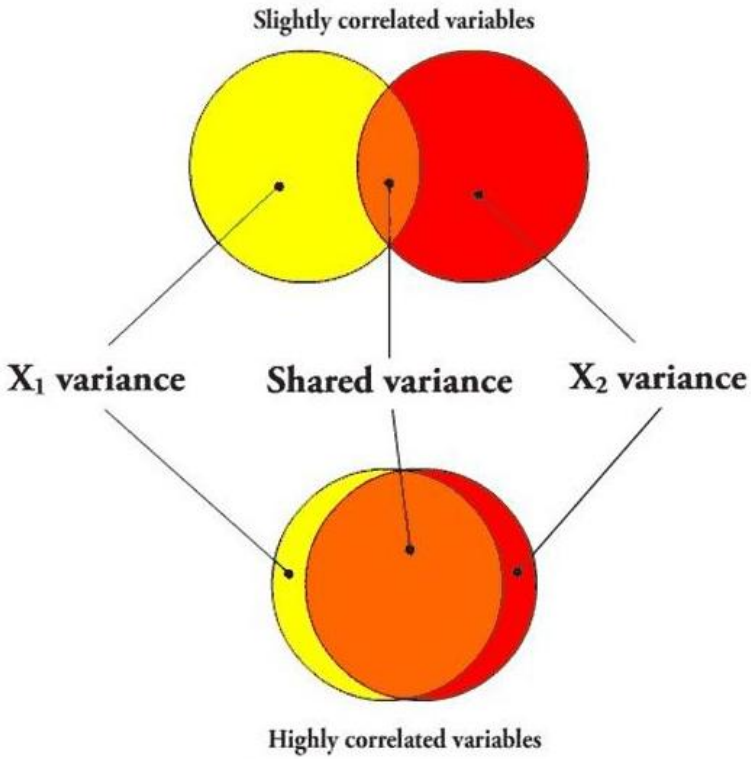
FIGURE 3.12: Scatterplots of Crime against Percent Urban, Single Parent, and Poverty with OLS Fitted Lines

What to do about Outliers?

- Inform readers of the issue
- Re-run the results dropping the outlier; see if this affects your results. Are they robust to dropping or including the outlier?
- Justify whether you should include or exclude the observation.
- run a quantile regression(uses median)

Multicollinearity

- Multicollinearity occurs when independent variables are not independent from each other
- (Autocorrelation occurs when the residuals are not independent from each other.)
- variables that are highly correlated. Does not cause bias if model is properly specified.(only affects standard errors)
- Variance of $\hat{\beta}_1$ higher than it otherwise would
- Shows through higher than expected r^2 or Variance inflation factor(VIF)
- Remember: We interpet each IV as the isolated effect of each IV (Holding others constant).



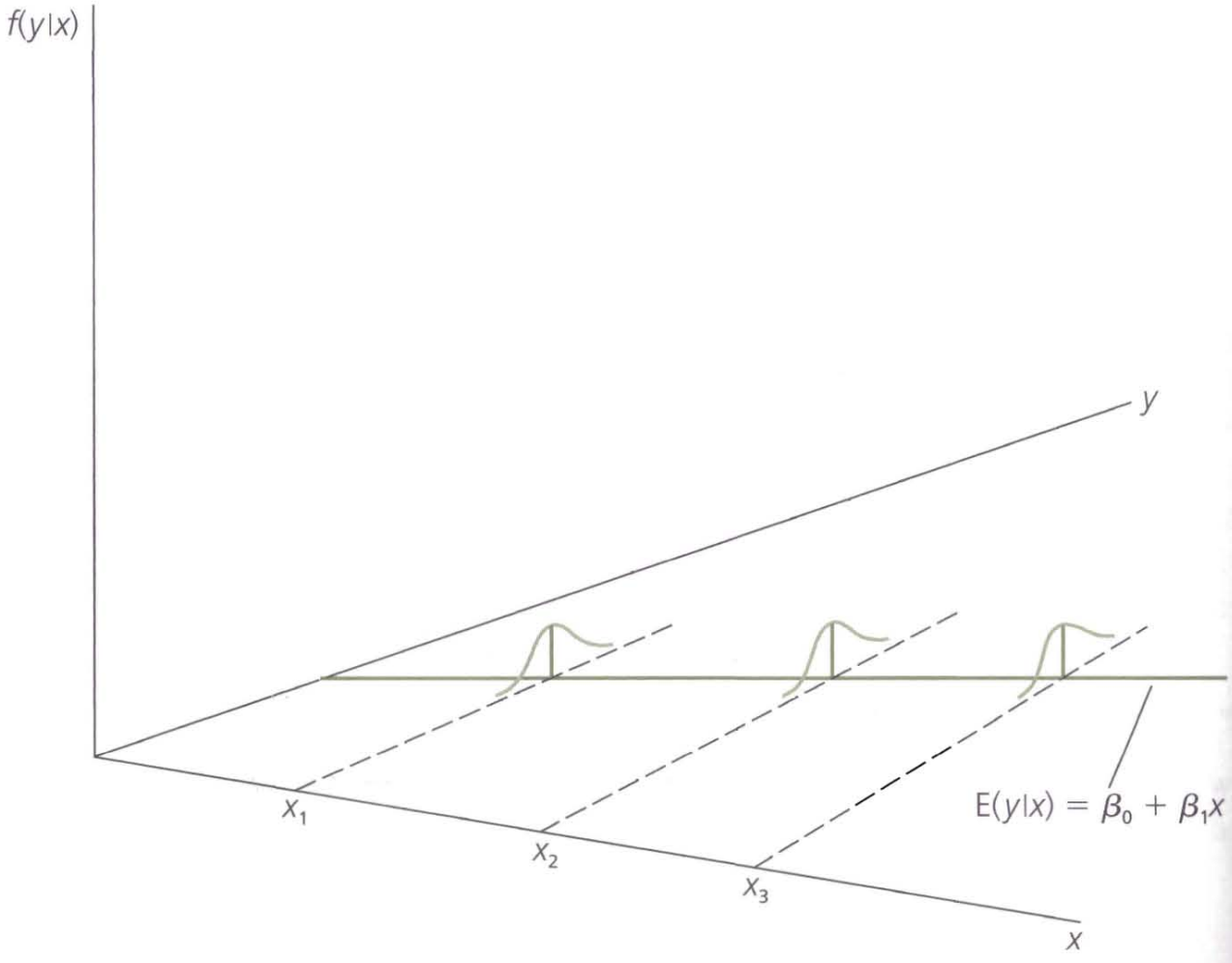
- Example: Years of Education and Highest Degree,

Issues with Errors: Homoskedasticity vs Heteroskedasiticity

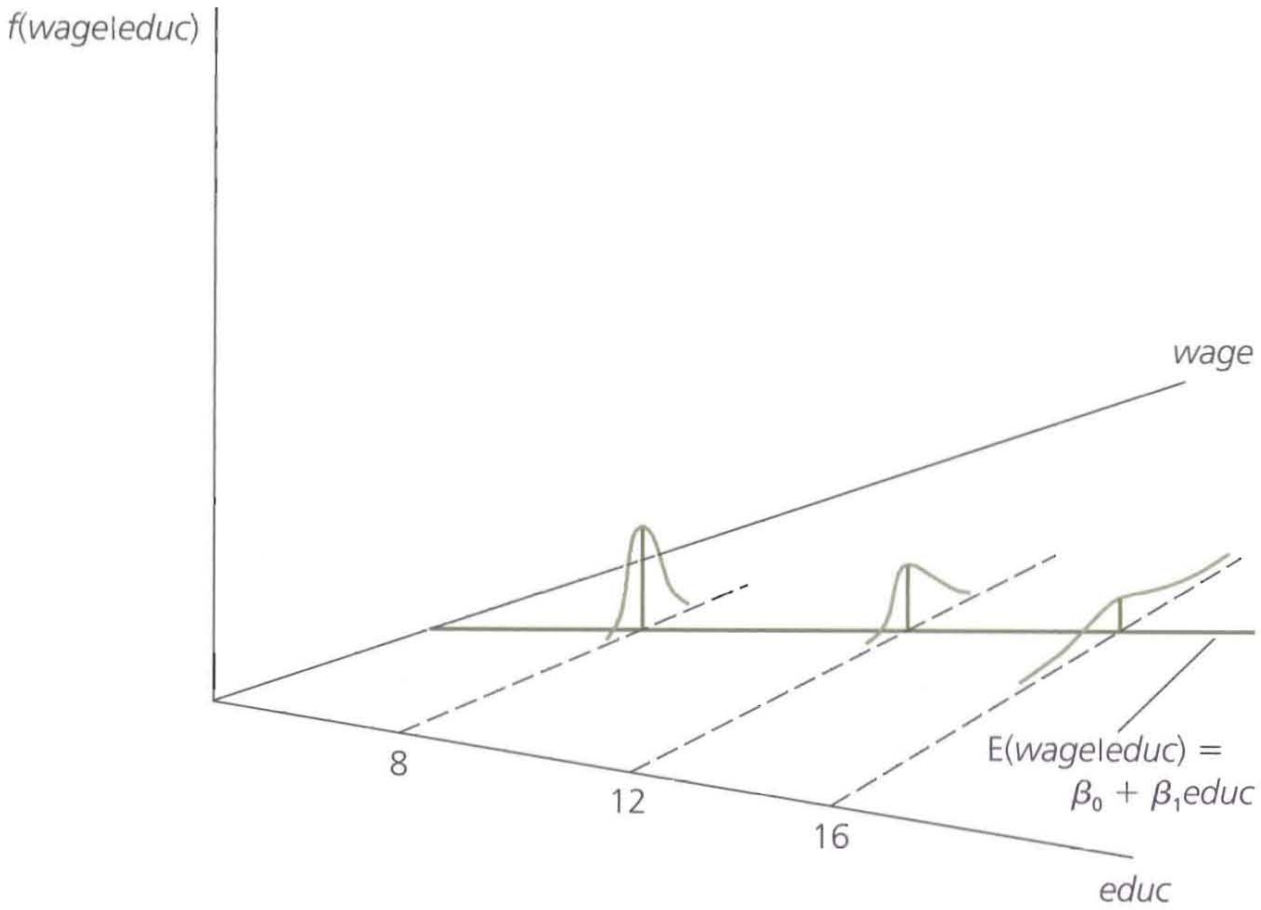
- Homoskedasticity: When the random variable, X, has the same variance for all observations of X. This isn't a problem.
- Heteroskedasticity: When the random variable, X, does not have the same variance for all observations of X. This is a problem, but fixable.

- **When we have heteroskedasticity, it means we have a non-constant variance in our errors.**

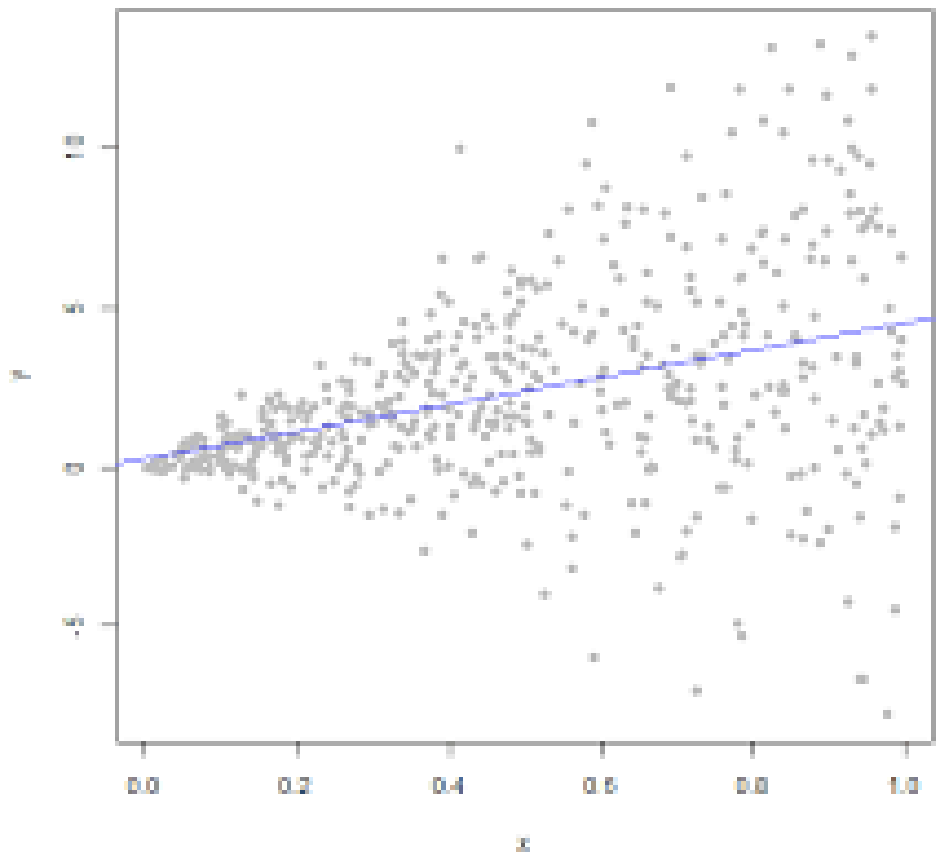
A Picture of Homoskedasticity



A Picture of Heteroskedasticity



A Picture of Heteroskedasticity



Fixing the Heteroskedasiticity Problem

- Heteroskedasticity: When the random variable, X, does not have the same variance for all observations of X. This is a fixable problem.
- Remember, this only affects our standard errors, σ, not our β_1 estimate. Therefore, it doesn't cause bias.
- So what's the solution? We use a slightly different estimator for our standard error calculations (e.g. White's heteroskedasticity-consistent estimator). Intuitively, we estimate the variance in our standard errors.
- Robust Standard Errors - You can do this in R.

Careful


Chelsea Parlett-Pelleriti @ChelseaParlett · Oct 14
PARENTS:
Make SURE you check your children's candy carefully this Halloween.
I just found HETEROSKEDASTICITY in this snickers bar😲😱
imgflip.com
70
858
6.3K

Time Series- Autocorrelation

Definition *Errors are* homoscedastic *if they have the same variance.*

Definition *Errors are* heteroscedastic *if they have different variance.*

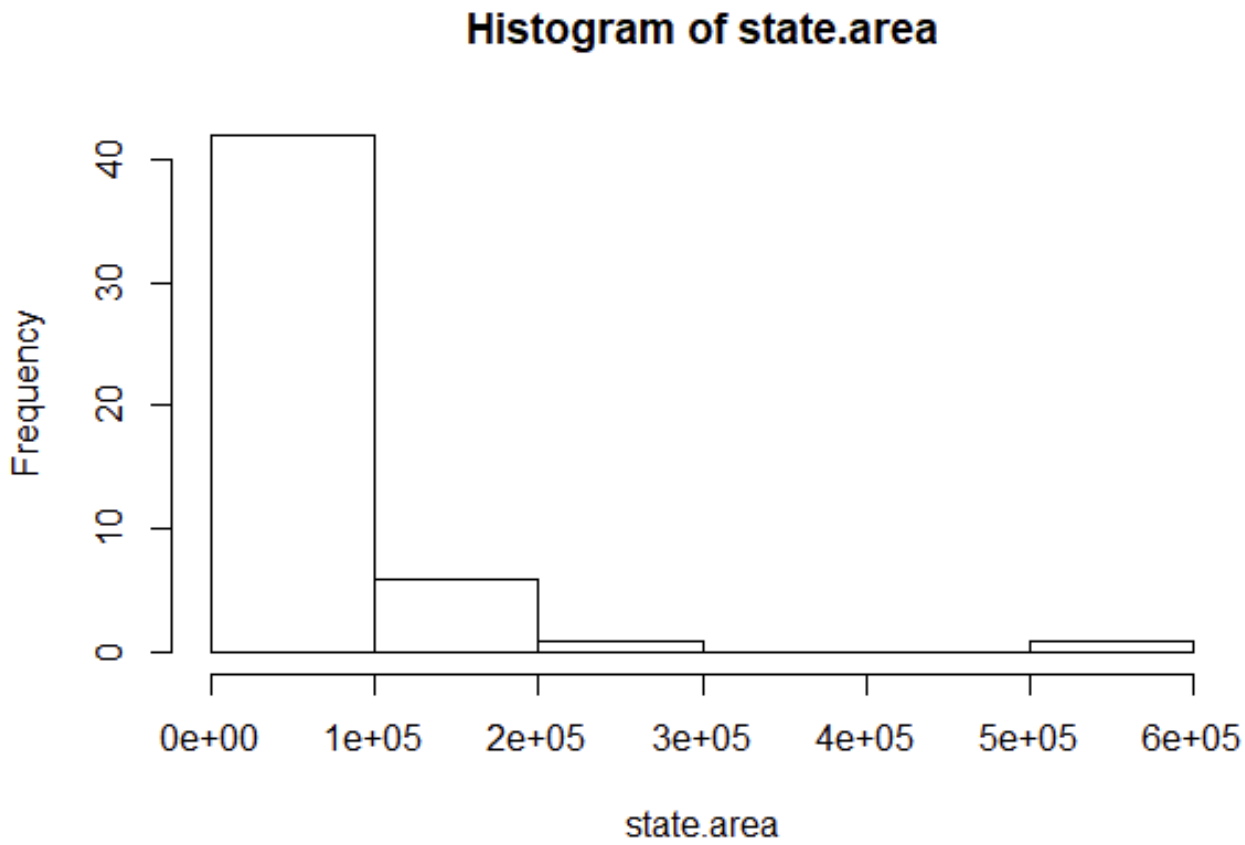
Definition *Errors are* autocorrelated *if the error from one observation is correlated with the error of another.*

- *If errors are autocorrelated knowing the error of observation 10 would provide information about the likely error of observation 11.*
- *Autocorrelation is common in time series data.*

Assume of Normality

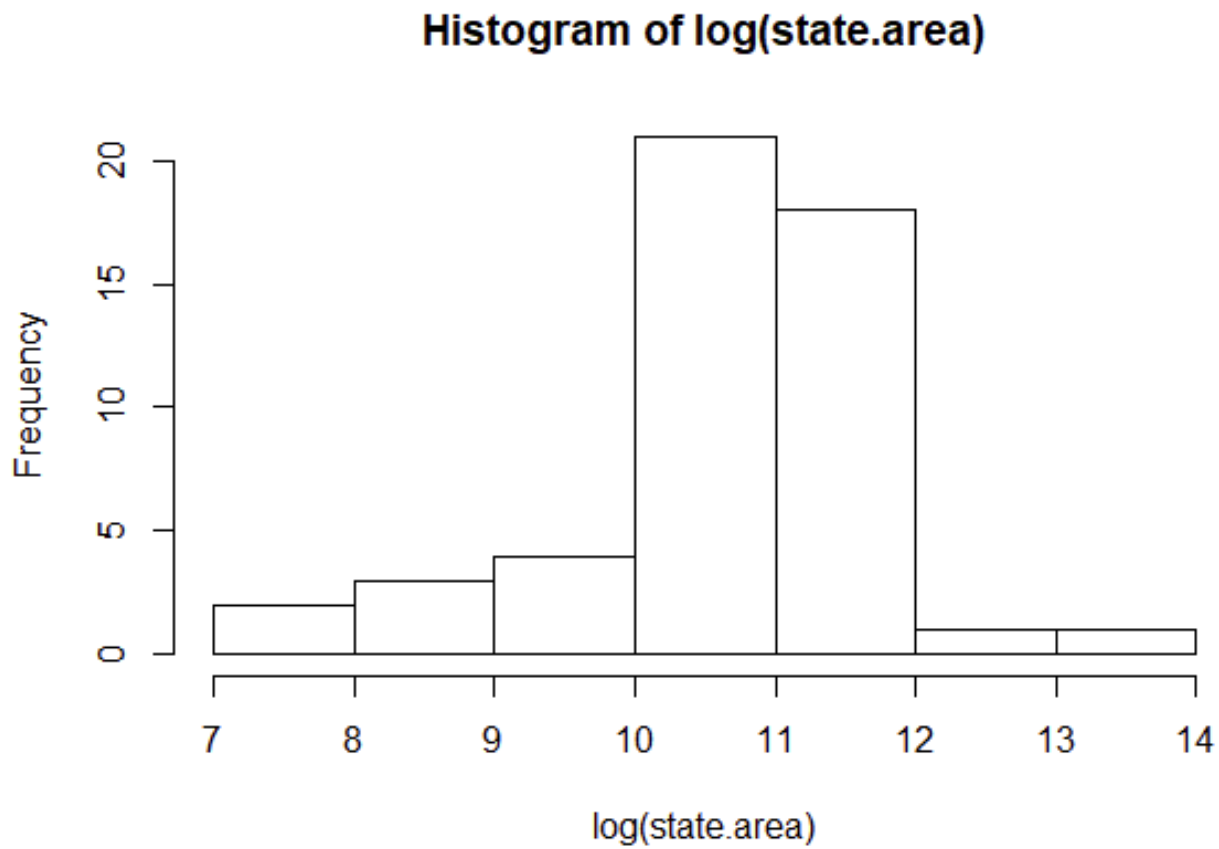
We often assume, for good reason, that our variables have a normal distribution However, in some cases the distribution is skewed. Ie.

We've seen this often with income data.



Assume of Normality

To make a skewed distribution we can “log” the variable



However, this complicates our interpretation.

Regression Warnings

Assumptions that can invalidate your results if they don't hold

1. Continuous DV
2. IV – continuous normally distributed or dummy
3. Linear relationship
4. No significant outliers
5. Independence of observations
6. Homoskedasticity
7. Residuals (errors) of regression line normally distributed

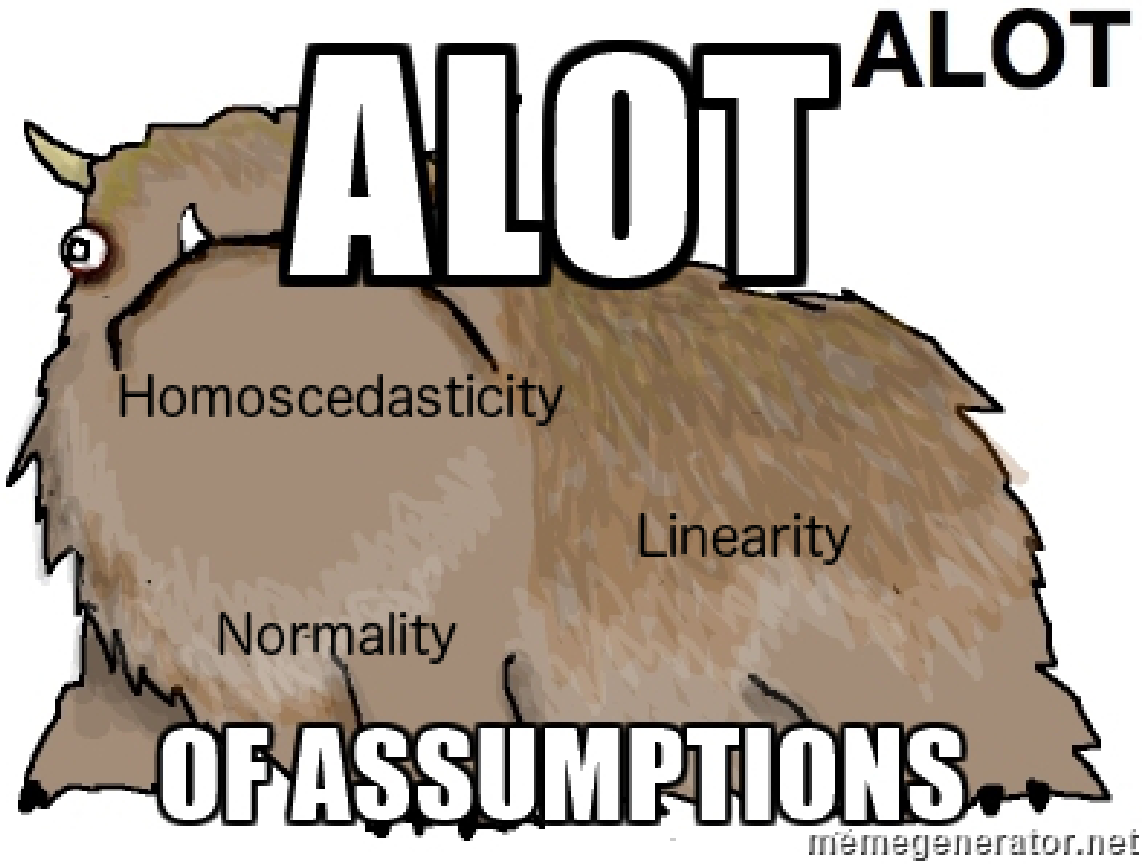
BLUE

Gauss-Markov Assumptions

- Best - lowest variance of the estimate
- Linear
- Unbiased - $\hat{\beta}_1 = \beta_1$
- Estimators



Regression is a very powerful tool when used correctly
However, when used incorrectly it can be dangerous


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Homoscedasticity
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Problems to be aware of

- Omitted Variable Bias
- Non-linearity
- Outliers
- Heteroskedasticity
- Autocorrelation
- Multicollinearity

What Breaks What?

Your coefficients are wrong *(bias)*	Omitted variables
	Wrong functional form
	Influential outliers
	Measurement error in X
Your standard errors are wrong *(inference)*	Heteroskedasticity
	Autocorrelation
	Clustering
	Multicollinearity
Mostly nothing	Non-normal errors (at our N)

Bias is a problem with your **model**. Standard errors are a problem with your **inference**.

Next Time: Experiments
Read: Bailey ch 10
Imbens and Rubin “Causality: The Basic Framework from Causal Inference”